



Customer Shopping Behavior Analysis

Transactional insights from **3,900 purchases** across product categories — uncovering spending patterns, segments, and strategic opportunities.

Dataset at a Glance

3,900

Transactions

18

Features

37

Missing Ratings

Customer Data

- Age, Gender, Location
- Subscription Status

Purchase Data

- Item, Category, Amount
- Discount, Shipping, Reviews

Data Preparation in Python

01

Load & Explore

pandas import, df.info(), describe()

02

Clean

Imputed Review Rating via category
median

03

Standardize

Renamed columns to snake_case

04

Engineer

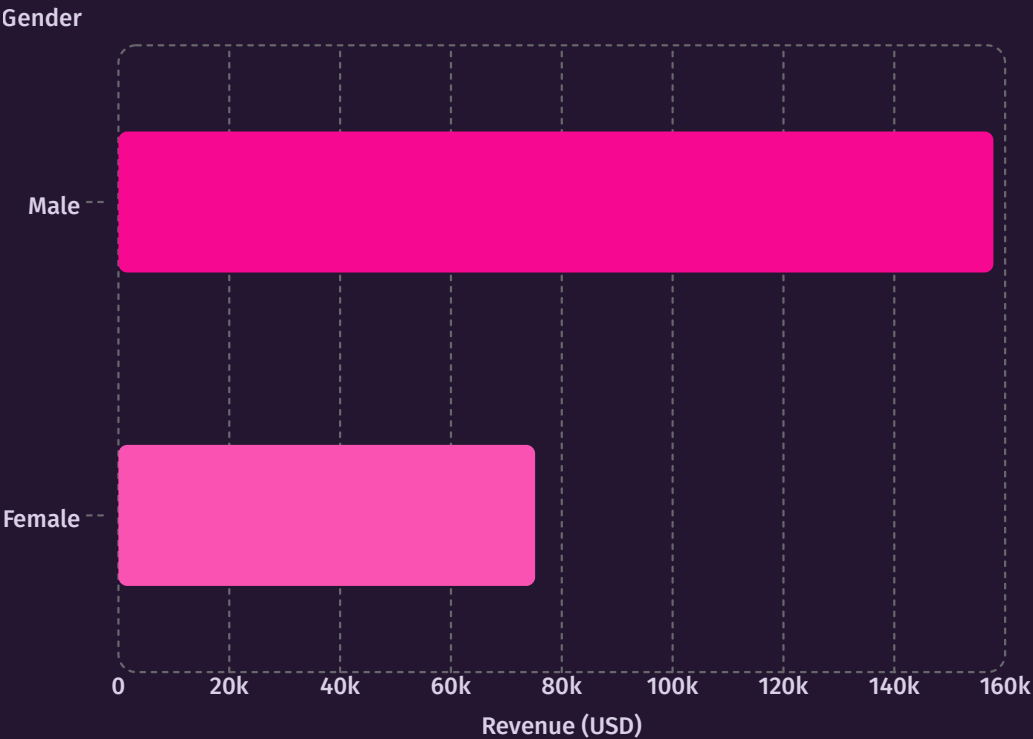
Created age_group & purchase_frequency_days

05

Load to DB

Connected to PostgreSQL for SQL analysis

Revenue by Gender



Key Insight

Male customers generated **2.1x more revenue** than female customers across 3,900 transactions.

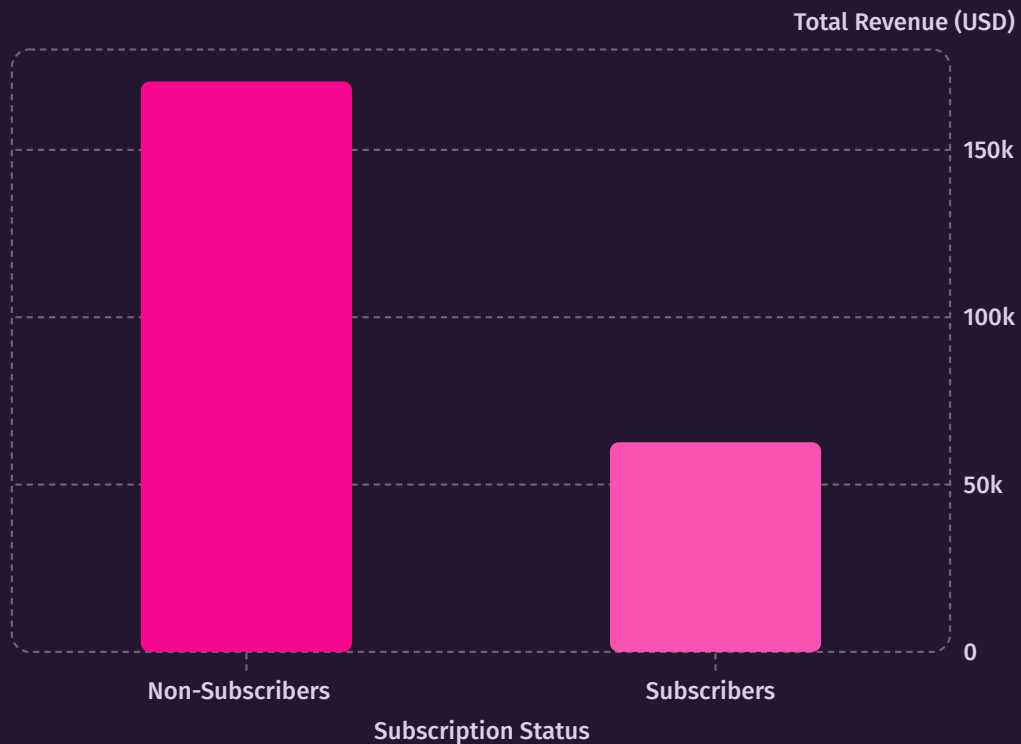
Male

\$157,890

Female

\$75,191

Subscribers vs. Non-Subscribers



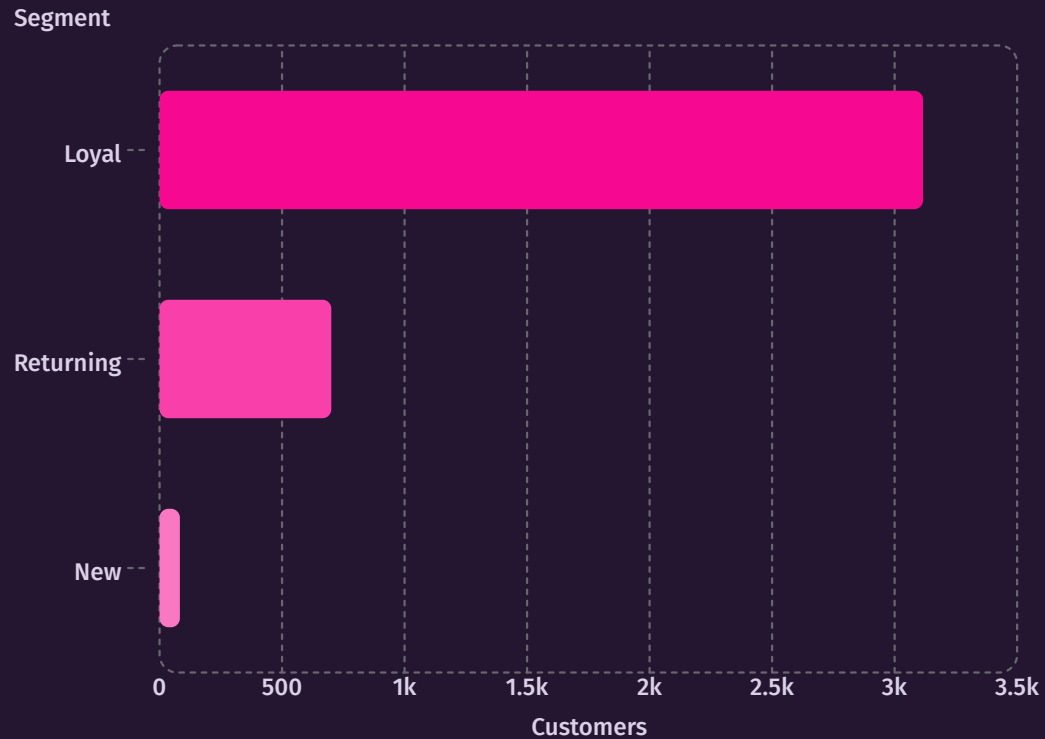
Comparison

Non-subscribers drive **73%** of total revenue despite similar average spend per transaction.

- Avg spend: ~\$59.50 (both groups)
- Non-subscribers: 2,847 customers
- Subscribers: 1,053 customers

📌 **Opportunity:** Convert non-subscribers to boost recurring revenue.

Customer Segmentation



Segment Breakdown

Loyal – 3,116

80% of customer base

Returning – 701

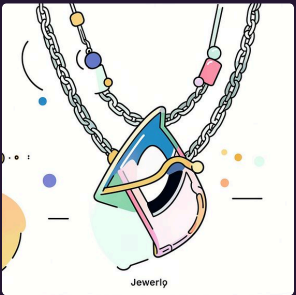
18% of customer base

New – 83

2% of customer base

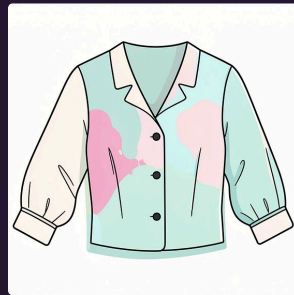
Top Products & Categories

Top Products per Category



Accessories

Jewelry, Sunglasses, Belt



Clothing

Blouse, Pants, Shirt



Footwear

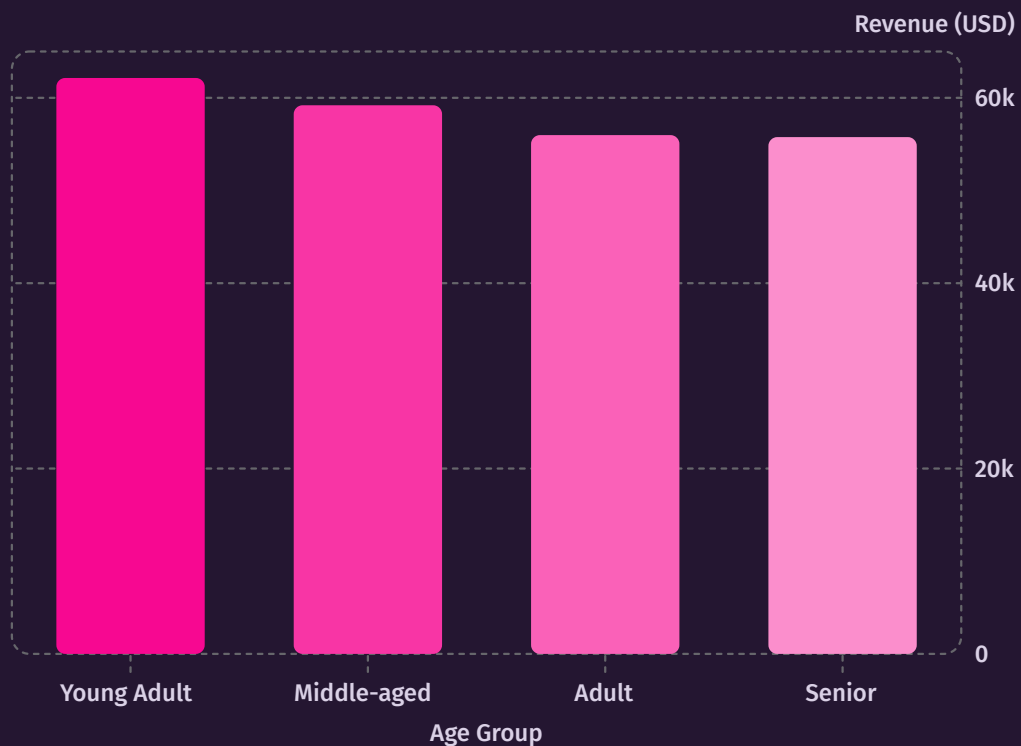
Sandals, Shoes, Sneakers



Outerwear

Jacket, Coat

Revenue by Age Group



Takeaway

Revenue is **evenly distributed** across age groups — Young Adults lead by a narrow margin.

- Young Adult: \$62,143
- Middle-aged: \$59,197
- Adult: \$55,978
- Senior: \$55,763

Power BI Dashboard



Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers to grow Loyal segment



Review Discount Policy

Balance sales lift with margin control



Targeted Marketing

Focus on high-revenue age groups & express-ship users